

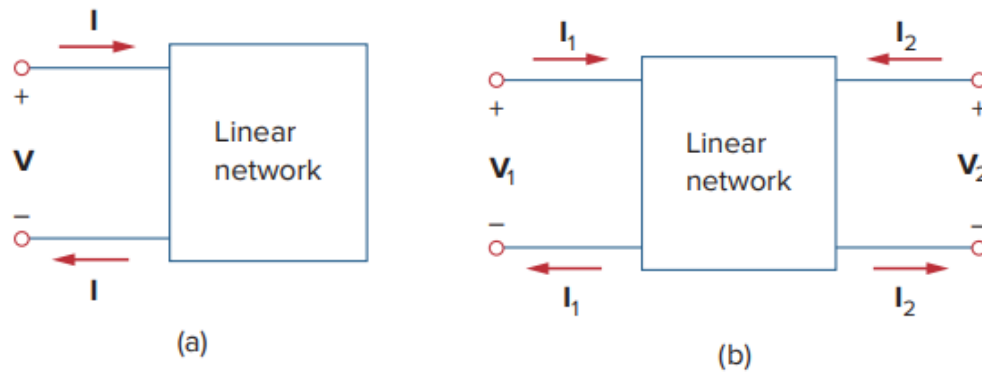
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Chapter 1

Two-port network

1.1 Definitions



1. **Port** = a pair of terminals through which current may enter or leave a network.
2. A **port** is an access point to the network and consists of a pair of terminals; the current entering one terminal leaves through the other terminal so that the net current entering the port equals zero.
3. (Left figure) A **one-port network** (two terminal circuit) has a single pair of terminals, such as the two terminals of a resistor, a capacitor, or an inductor.
4. (Right figure) **Two-port network** is an electrical network with two separate ports for input and output.
5. Thus, a two-port network has two terminal pairs acting as access points. For example, three-terminal devices, such as transistors, can be configured into two-port networks.
6. Two reasons to study two-port networks
 - (a) Two-port networks are useful in communications, control systems, power systems, and electronics. For example, they are used in electronics to model transistors and to facilitate cascaded design.
 - (b) Knowing the parameters of a two-port network enables us to treat it as a "black-box" when embedded within a larger network.
7. Characterization of two-port networks
 - (a) **Terminal quantities** : V_1, V_2, I_1, I_2 , out of which two are independent.
 - (b) **Parameters** : Various terms that relate to these voltages and currents

1.2 Impedance and Admittance parameters

1.2.1 Impedance (z) parameters

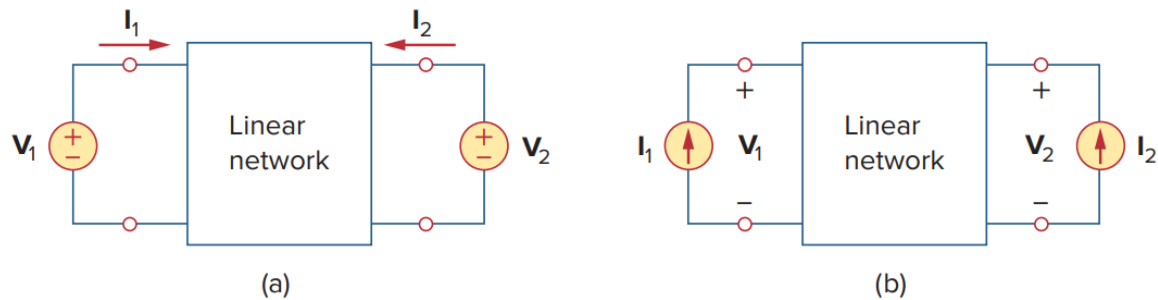


Figure 19.2

The linear two-port network: (a) driven by voltage sources, (b) driven by current sources.

1. **Impedance parameters** relate the terminal currents $\mathbf{I}_1, \mathbf{I}_2$ (independent terminal quantities) to the terminal voltages $\mathbf{V}_1, \mathbf{V}_2$ (dependent terminal quantities) as

$$\mathbf{V}_1 = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2$$

$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2$$

or in matrix form as

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{z}_{11} & \mathbf{z}_{12} \\ \mathbf{z}_{21} & \mathbf{z}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = [\mathbf{z}] \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix}$$

Here, the subscripts of a $\mathbf{z}$ parameter $\mathbf{z}_{ab}$ denote

a : output terminal

b : input terminal

For example, $\mathbf{z}_{12}$ is a measure of how much output voltage $\mathbf{V}_1$ is produced for an input current $\mathbf{I}_2$.

2. Impedance parameters do not exist when the terminal quantities cannot be described as the above matrix equation. For example, the ideal transformer has no z parameters

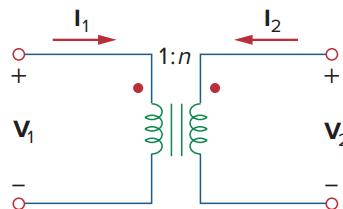


Figure 19.6

An ideal transformer has no z parameters.

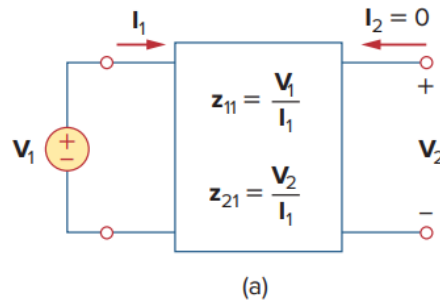
because the terminal quantities are related as

$$\mathbf{V}_1 = \frac{1}{n}\mathbf{V}_2, \quad \mathbf{I}_1 = -n\mathbf{I}_2$$

However, the ideal transformer has hybrid parameters.

3. **Determination of z-parameters.** Here, port 1 is designated as the input port, while port 2 is designated as the output port.

- (a) Evaluation of $\mathbf{z}_{11}, \mathbf{z}_{21}$: apply an input terminal current $\mathbf{I}_1$ while open-circuiting the output-port (the terminal current $\mathbf{I}_2 = 0$)

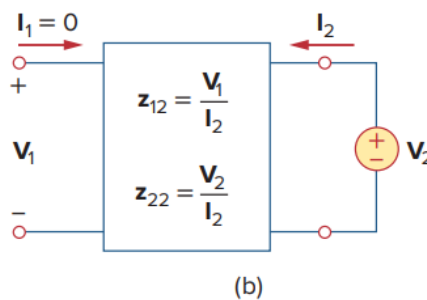


$$\mathbf{z}_{11} = \left. \frac{\mathbf{V}_1}{\mathbf{I}_1} \right|_{\mathbf{I}_2=0}, \quad \mathbf{z}_{21} = \left. \frac{\mathbf{V}_2}{\mathbf{I}_1} \right|_{\mathbf{I}_2=0}$$

For this reason, we define each $\mathbf{z}$ parameter as

- i. $\mathbf{z}_{11}$ = Open-circuit input impedance
- ii. $\mathbf{z}_{21}$ = Open-circuit transfer impedance from port 1 to port 2

- (b) Evaluation of $\mathbf{z}_{12}, \mathbf{z}_{22}$: apply an output terminal current $\mathbf{I}_2$ while open-circuiting the input-port (the terminal current $\mathbf{I}_1 = 0$)



$$\mathbf{z}_{12} = \left. \frac{\mathbf{V}_1}{\mathbf{I}_2} \right|_{\mathbf{I}_1=0}, \quad \mathbf{z}_{22} = \left. \frac{\mathbf{V}_2}{\mathbf{I}_2} \right|_{\mathbf{I}_1=0}$$

For this reason, we define each $\mathbf{z}$ parameter as

- i. $\mathbf{z}_{12}$ = Open-circuit output impedance
- ii. $\mathbf{z}_{22}$ = Open-circuit transfer impedance from port 2 to port 1

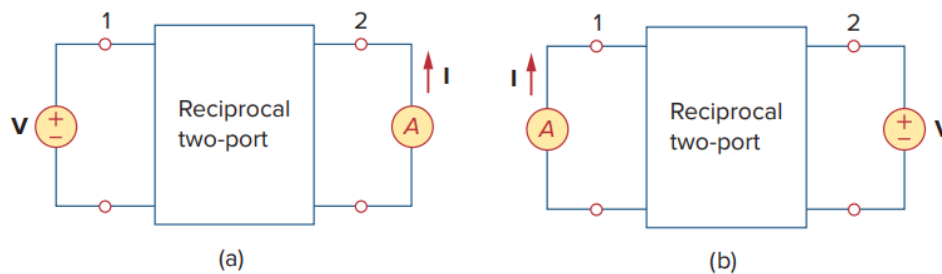
- (c) Sometimes $\mathbf{z}_{11}$ and $\mathbf{z}_{22}$ are called driving-point impedances, the input impedance of a two-terminal(one-port) device.

4. Special cases of the two-port networks

- (a) **Symmetrical** two-port network : when $\mathbf{z}_{11} = \mathbf{z}_{22}$.
- (b) **Reciprocal** two-port network : if the two-port network
 - i. is linear
 - ii. has no dependent sources
 then transfer impedances are equal,

$$\mathbf{z}_{12} = \mathbf{z}_{21}$$

In other words, if the points of excitation and response are interchanged, the transfer impedances remain the same. This can be proved as follows.



As illustrated in the above figure, a two-port is reciprocal if interchanging an ideal voltage source at one port with an ideal ammeter at the other port gives the same ammeter reading.

When the ideal voltage source $\mathbf{V}$ is connected to the input port as in Figure (a), the ammeter reading at the output port is

$$\mathbf{I}_1 = \frac{\mathbf{V}}{\mathbf{z}_{12}}$$

On the other hand, when the ideal voltage source $\mathbf{V}$ is connected to the output port as in Figure (b), the ammeter reading at the input port is

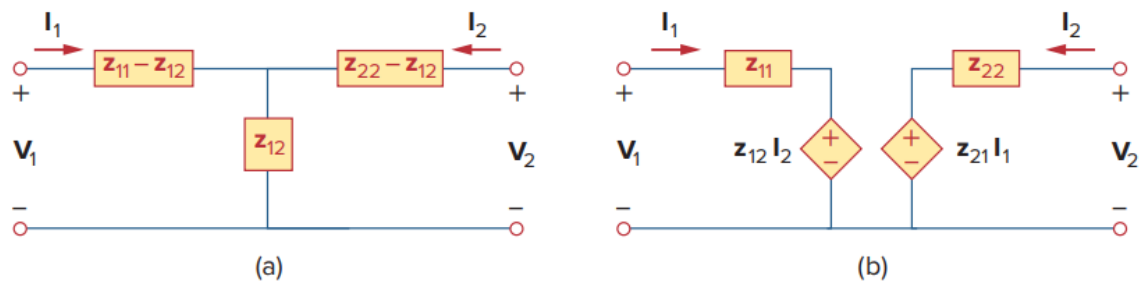
$$\mathbf{I}_2 = \frac{\mathbf{V}}{\mathbf{z}_{21}}$$

Then $\mathbf{I}_1 = \mathbf{I}_2$ is possible only if this two-port network is reciprocal,

$$\mathbf{z}_{12} = \mathbf{z}_{21}.$$

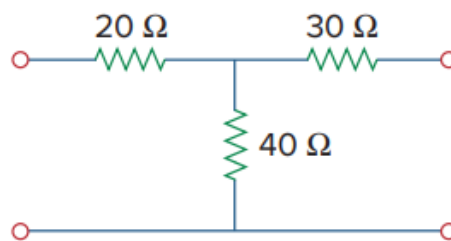
Remark. Any two-port that is made entirely of resistors, capacitors, and inductors must be reciprocal.

5. Equivalent circuits of two-port networks

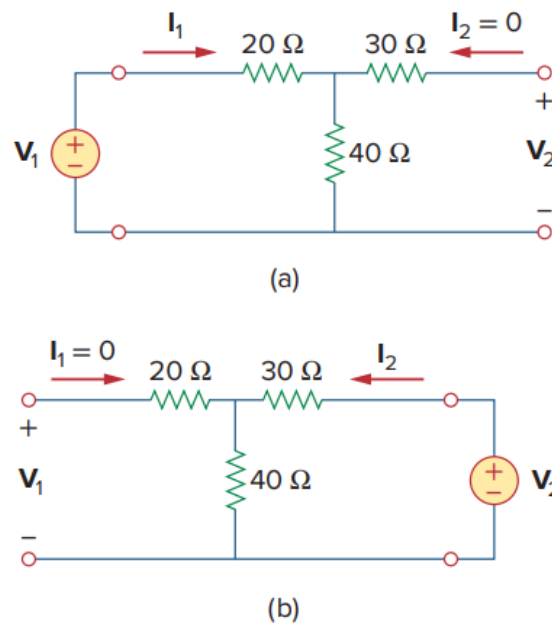
**Figure 19.5**

(a) T-equivalent circuit (for reciprocal case only), (b) general equivalent circuit.

6. **Example.** Determine the z parameters for the circuit in Fig. 19.7.

**Figure 19.7**

For Example 19.1.

Solution 1. Open-circuiting each port**Figure 19.8**

For Example 19.1: (a) finding $\mathbf{z}_{11}$ and $\mathbf{z}_{21}$,
 (b) finding $\mathbf{z}_{12}$ and $\mathbf{z}_{22}$.

To determine $\mathbf{z}_{11}$ and $\mathbf{z}_{21}$, we apply a voltage source $\mathbf{V}_1$ to the input port and leave the output port open as in Fig. 19.8(a). Then,

$$\mathbf{z}_{11} = \frac{\mathbf{V}_1}{\mathbf{I}_1} = \frac{(20 + 40)\mathbf{I}_1}{\mathbf{I}_1} = 60 \, \Omega$$

that is, $\mathbf{z}_{11}$ is the input impedance at port 1.

$$\mathbf{z}_{21} = \frac{\mathbf{V}_2}{\mathbf{I}_1} = \frac{40\mathbf{I}_1}{\mathbf{I}_1} = 40 \, \Omega$$

To find $\mathbf{z}_{12}$ and $\mathbf{z}_{22}$, we apply a voltage source $\mathbf{V}_2$ to the output port and leave the input port open as in Fig. 19.8(b). Then,

$$\mathbf{z}_{12} = \frac{\mathbf{V}_1}{\mathbf{I}_2} = \frac{40\mathbf{I}_2}{\mathbf{I}_2} = 40 \, \Omega, \quad \mathbf{z}_{22} = \frac{\mathbf{V}_2}{\mathbf{I}_2} = \frac{(30 + 40)\mathbf{I}_2}{\mathbf{I}_2} = 70 \, \Omega$$

Thus,

$$[\mathbf{z}] = \begin{bmatrix} 60 \, \Omega & 40 \, \Omega \\ 40 \, \Omega & 70 \, \Omega \end{bmatrix}$$

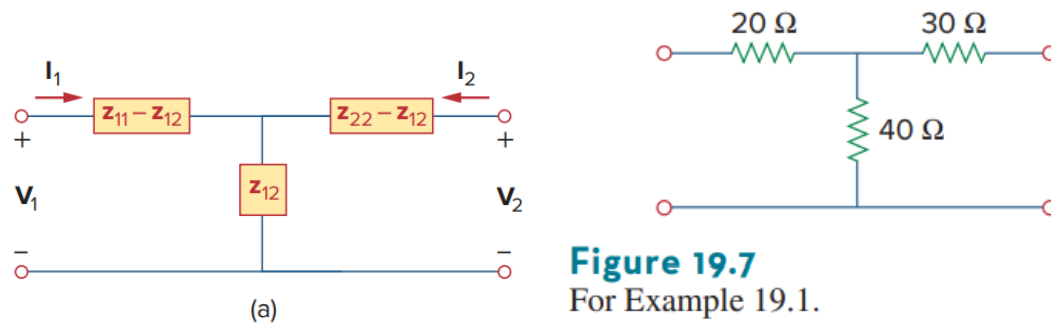
Solution 2. T-equivalent circuit

Figure 19.7
For Example 19.1.

Alternatively, as there is no dependent source in the given circuit, which only consists of resistors (linear circuit elements), $z_{12} = z_{21}$ and we can use T-equivalent circuit of a reciprocal network. Comparing Fig. 19.7 with the T-equivalent circuit above, we get

$$z_{12} = 40\ \Omega = z_{21}$$

$$z_{11} - z_{12} = 20 \quad \Rightarrow \quad z_{11} = 20 + z_{12} = 60\ \Omega$$

$$z_{22} - z_{12} = 30 \quad \Rightarrow \quad z_{22} = 30 + z_{12} = 70\ \Omega$$

1.2.2 Admittance parameters

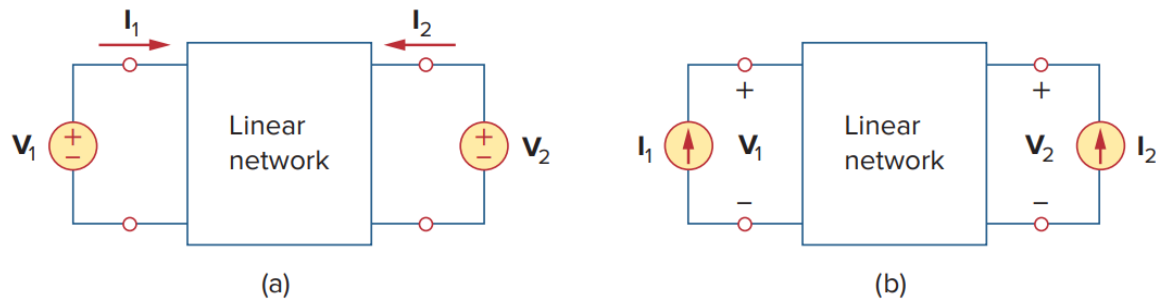


Figure 19.2

The linear two-port network: (a) driven by voltage sources, (b) driven by current sources.

1. **Admittance parameters** relate the terminal currents $\mathbf{I}_1, \mathbf{I}_2$ (independent terminal quantities) to the terminal voltages $\mathbf{V}_1, \mathbf{V}_2$ (dependent terminal quantities) as

$$\mathbf{I}_1 = \mathbf{y}_{11} \mathbf{V}_1 + \mathbf{y}_{12} \mathbf{V}_2$$

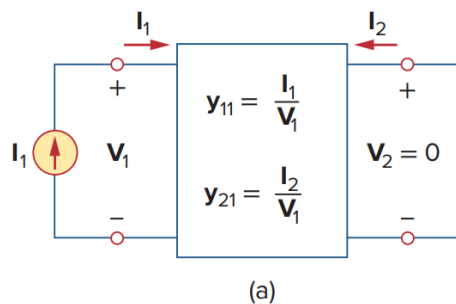
$$\mathbf{I}_2 = \mathbf{y}_{21} \mathbf{V}_1 + \mathbf{y}_{22} \mathbf{V}_2$$

or in matrix form as

$$\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{y}_{11} & \mathbf{y}_{12} \\ \mathbf{y}_{21} & \mathbf{y}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = [\mathbf{y}] \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix}$$

2. **Determination of $\mathbf{y}$ parameters** Here, port 1 is designated as the input port, while port 2 is designated as the output port.

- (a) Evaluation of $\mathbf{y}_{11}, \mathbf{y}_{21}$: apply an input terminal voltage $\mathbf{V}_1$ while short-circuiting the output-port (the terminal voltage $\mathbf{V}_2 = 0$)

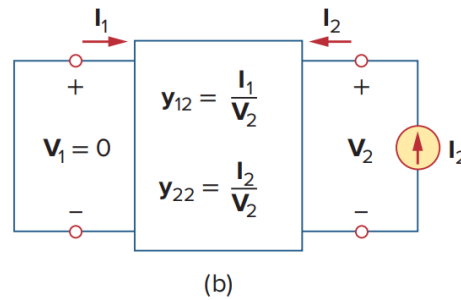


$$\mathbf{y}_{11} = \left. \frac{\mathbf{I}_1}{\mathbf{V}_1} \right|_{\mathbf{V}_2=0}, \quad \mathbf{y}_{21} = \left. \frac{\mathbf{I}_2}{\mathbf{V}_1} \right|_{\mathbf{V}_2=0}$$

For this reason, we define each $\mathbf{y}$ parameter as

- i. $\mathbf{y}_{11}$ = Short-circuit input admittance
- ii. $\mathbf{y}_{21}$ = Short-circuit transfer admittance from port 1 to port 2

- (b) Evaluation of $\mathbf{y}_{12}, \mathbf{y}_{22}$: apply an output terminal voltage $\mathbf{V}_2$ while short-circuiting the input-port (the terminal voltage $\mathbf{V}_1 = 0$)



$$\mathbf{y}_{12} = \left. \frac{\mathbf{I}_1}{\mathbf{V}_2} \right|_{\mathbf{V}_1=0}, \quad \mathbf{y}_{22} = \left. \frac{\mathbf{I}_2}{\mathbf{V}_2} \right|_{\mathbf{V}_1=0}$$

For this reason, we define each $\mathbf{y}$ parameter as

- i. $\mathbf{y}_{12}$ = Short-circuit output admittance
- ii. $\mathbf{y}_{22}$ = Short-circuit transfer admittance from port 2 to port 1

3. Reciprocal two-port network : if the two-port network

- (a) is linear
- (b) has no dependent sources

then transfer admittances are equal,

$$\mathbf{y}_{12} = \mathbf{y}_{21}$$

A reciprocal network can be modeled by the Π -equivalent circuit in Fig. 19.13(a). If the network is not reciprocal, a more general equivalent network is shown in Fig. 19.13(b).

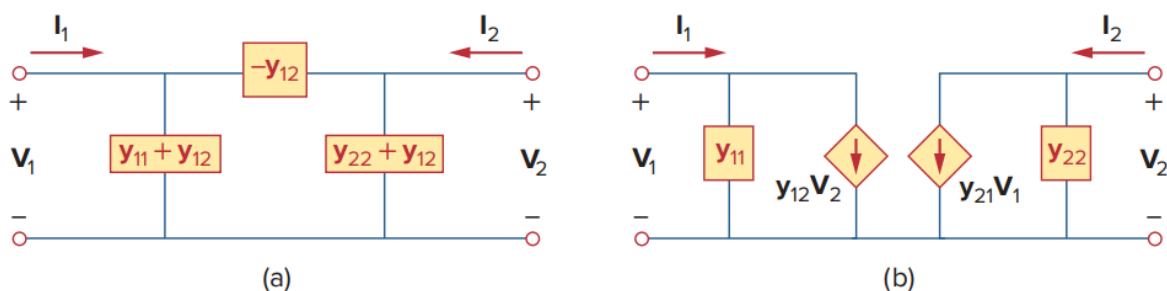


Figure 19.13

(a) Π -equivalent circuit (for reciprocal case only), (b) general equivalent circuit.

4. **Example.** Obtain the $\mathbf{y}$ parameters for the Π network shown in Fig. 19.14.

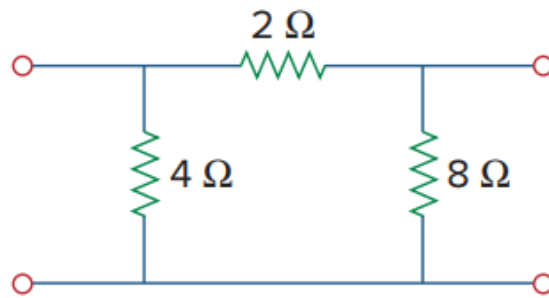
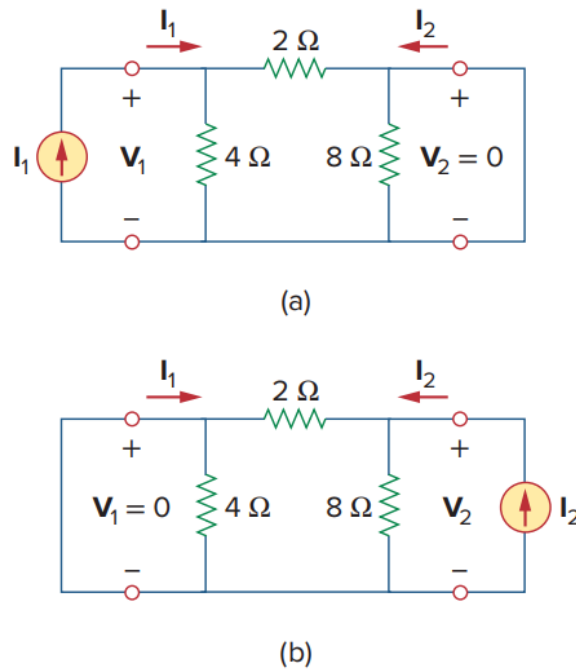


Figure 19.14
For Example 19.3.

Solution 1. Short-circuiting ports**Figure 19.15**

For Example 19.3: (a) finding $\mathbf{y}_{11}$ and $\mathbf{y}_{21}$,
 (b) finding $\mathbf{y}_{12}$ and $\mathbf{y}_{22}$.

To find $\mathbf{y}_{11}$ and $\mathbf{y}_{21}$, short-circuit the output port and connect a current source $\mathbf{I}_1$ to the input port as in Fig. 19.15(a). Because the $8\ \Omega$ resistor is short-circuited, the $2\ \Omega$ resistor is in parallel with the $4\ \Omega$ resistor. Hence,

$$\mathbf{V}_1 = \mathbf{I}_1(4 \parallel 2) = \frac{4}{3}\mathbf{I}_1, \quad \mathbf{y}_{11} = \frac{\mathbf{I}_1}{\mathbf{V}_1} = \frac{\mathbf{I}_1}{\frac{4}{3}\mathbf{I}_1} = 0.75\ \text{S}$$

By current division,

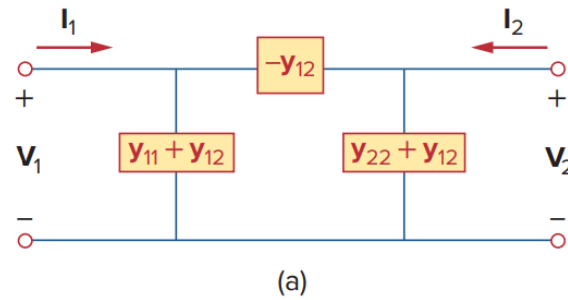
$$-\mathbf{I}_2 = \frac{4}{4+2}\mathbf{I}_1 = \frac{2}{3}\mathbf{I}_1, \quad \mathbf{y}_{21} = \frac{\mathbf{I}_2}{\mathbf{V}_1} = \frac{-\frac{2}{3}\mathbf{I}_1}{\frac{4}{3}\mathbf{I}_1} = -0.5\ \text{S}$$

To get $\mathbf{y}_{12}$ and $\mathbf{y}_{22}$, short-circuit the input port and connect a current source $\mathbf{I}_2$ to the output port as in Fig. 19.15(b). The $4\ \Omega$ resistor is short-circuited so that the 2- and $8\ \Omega$ resistors are in parallel.

$$\mathbf{V}_2 = \mathbf{I}_2(8 \parallel 2) = \frac{8}{5}\mathbf{I}_2, \quad \mathbf{y}_{22} = \frac{\mathbf{I}_2}{\mathbf{V}_2} = \frac{\mathbf{I}_2}{\frac{8}{5}\mathbf{I}_2} = \frac{5}{8} = 0.625\ \text{S}$$

By current division,

$$-\mathbf{I}_1 = \frac{8}{8+2}\mathbf{I}_2 = \frac{4}{5}\mathbf{I}_2, \quad \mathbf{y}_{12} = \frac{\mathbf{I}_1}{\mathbf{V}_2} = \frac{-\frac{4}{5}\mathbf{I}_2}{\frac{8}{5}\mathbf{I}_2} = -0.5\ \text{S}$$

Solution 2. Π network

Alternatively, comparing Fig. 19.14 with Fig. 19.13(a),

$$\mathbf{y}_{12} = -\frac{1}{2} \mathbf{S} = \mathbf{y}_{21}$$

$$\mathbf{y}_{11} + \mathbf{y}_{12} = \frac{1}{4} \quad \Rightarrow \quad \mathbf{y}_{11} = \frac{1}{4} - \mathbf{y}_{12} = 0.75 \text{ S}$$

$$\mathbf{y}_{22} + \mathbf{y}_{12} = \frac{1}{8} \quad \Rightarrow \quad \mathbf{y}_{22} = \frac{1}{8} - \mathbf{y}_{12} = 0.625 \text{ S}$$

5. **Example.** Determine the y parameters for the two-port shown in Fig. 19.17.

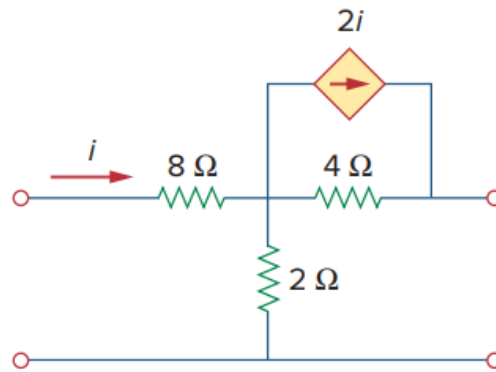
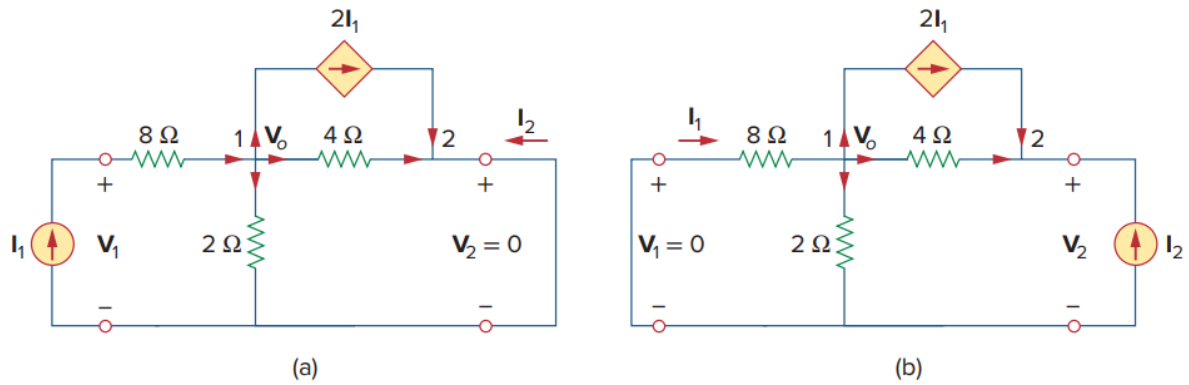


Figure 19.17
For Example 19.4.

**Figure 19.18**

Solution of Example 19.4: (a) finding y_{11} and y_{21} , (b) finding y_{12} and y_{22} .

Solution: We follow the same procedure as in the previous example. To get y_{11} and y_{21} , we use the circuit in Fig. 19.18(a), in which port 2 is short-circuited and a current source is applied to port 1.

At node 1,

$$\frac{V_1 - V_o}{8} = 2I_1 + \frac{V_o}{2} + \frac{V_o - 0}{4}$$

But

$$I_1 = \frac{V_1 - V_o}{8};$$

therefore,

$$0 = \frac{V_1 - V_o}{8} + \frac{3V_o}{4}$$

$$0 = V_1 - V_o + 6V_o \Rightarrow V_1 = -5V_o$$

Hence,

$$I_1 = \frac{-5V_o - V_o}{8} = -0.75V_o$$

and

$$y_{11} = \frac{I_1}{V_1} = \frac{-0.75V_o}{-5V_o} = 0.15 \text{ S}$$

At node 2,

$$\frac{V_o - 0}{4} + 2I_1 + I_2 = 0$$

or

$$-I_2 = 0.25V_o - 1.5V_o = -1.25V_o$$

Hence,

$$y_{21} = \frac{I_2}{V_1} = \frac{1.25V_o}{-5V_o} = -0.25 \text{ S}$$

—

Similarly, we get y_{12} and y_{22} using Fig. 19.18(b).

At node 1,

$$\frac{0 - V_o}{8} = 2I_1 + \frac{V_o}{2} + \frac{V_o - V_2}{4}$$

But

$$\mathbf{I}_1 = \frac{0 - V_o}{8};$$

therefore,

$$0 = \frac{-V_o}{8} + \frac{V_o}{2} + \frac{V_o - \mathbf{V}_2}{4}$$

or

$$0 = -V_o + 4V_o + 2V_o - 2\mathbf{V}_2 \quad \Rightarrow \quad \mathbf{V}_2 = 2.5V_o$$

Hence,

$$\mathbf{y}_{12} = \frac{\mathbf{I}_1}{\mathbf{V}_2} = \frac{-V_o/8}{2.5V_o} = -0.05 \text{ S}$$

At node 2,

$$\frac{V_o - \mathbf{V}_2}{4} + 2\mathbf{I}_1 + \mathbf{I}_2 = 0$$

or

$$-\mathbf{I}_2 = 0.25V_o - \frac{1}{4}(2.5V_o) - \frac{2V_o}{8} = -0.625V_o$$

Thus,

$$\mathbf{y}_{22} = \frac{\mathbf{I}_2}{\mathbf{V}_2} = \frac{0.625V_o}{2.5V_o} = 0.25 \text{ S}$$

—

Notice that $\mathbf{y}_{12} \neq \mathbf{y}_{21}$ in this case, given that the network is not reciprocal

1.3 Hybrid parameters

1. **Hybrid (h) parameters** relate the terminal currents $\mathbf{I}_1, \mathbf{V}_2$ (independent terminal quantities) to the terminal voltages $\mathbf{V}_1, \mathbf{I}_2$ (dependent terminal quantities) as

$$\mathbf{V}_1 = h_{11}\mathbf{I}_1 + h_{12}\mathbf{V}_2$$

$$\mathbf{I}_2 = h_{21}\mathbf{I}_1 + h_{22}\mathbf{V}_2$$

or in matrix form as

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{V}_2 \end{bmatrix} = [h] \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{V}_2 \end{bmatrix}$$

2. They are very useful for describing **electronic** devices such as transistors (see Section 19.9); it is much easier to measure experimentally the h parameters of such devices than to measure their z or y parameters.
3. **Determination of h parameters** h_{11} and h_{21} are determined by short-circuiting output port ($\mathbf{V}_2 = 0$)

$$h_{11} = \left. \frac{\mathbf{V}_1}{\mathbf{I}_1} \right|_{\mathbf{V}_2=0}, \quad h_{21} = \left. \frac{\mathbf{I}_2}{\mathbf{I}_1} \right|_{\mathbf{V}_2=0},$$

h_{12} and h_{22} are determined by open-circuiting input port ($\mathbf{I}_1 = 0$)

$$h_{12} = \left. \frac{\mathbf{V}_1}{\mathbf{V}_2} \right|_{\mathbf{I}_1=0}, \quad h_{22} = \left. \frac{\mathbf{I}_2}{\mathbf{V}_2} \right|_{\mathbf{I}_1=0}$$

4. **Reciprocal** network condition : $h_{12} = -h_{21}$

5. **Equivalent circuit**

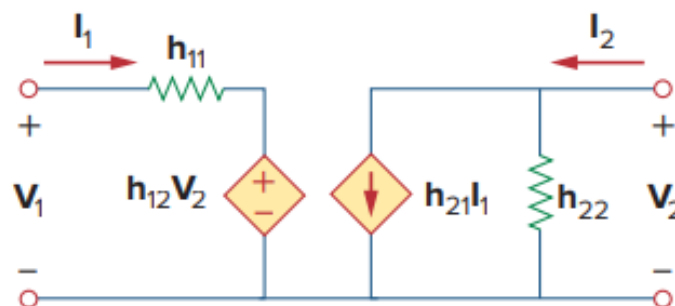


Figure 19.20

The h -parameter equivalent network of a two-port network.

6. **Example.** Find the hybrid parameters for the two-port network of Fig. 19.22.

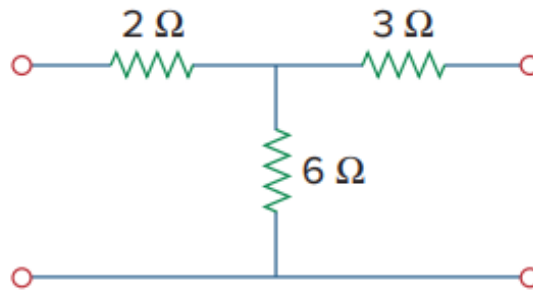


Figure 19.22
For Example 19.5.

Solution :

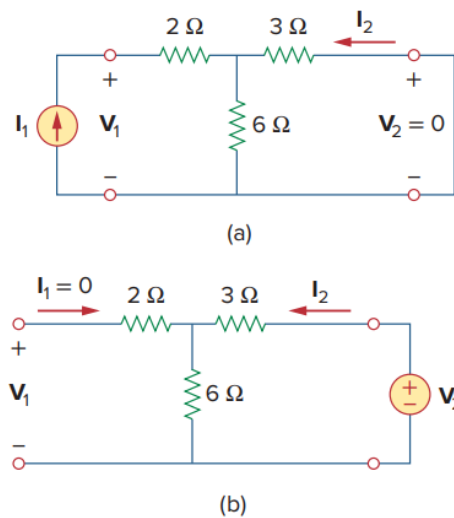


Figure 19.23

For Example 19.5: (a) computing h_{11} and h_{21} , (b) computing h_{12} and h_{22} .

To find h_{11} and h_{21} , we short-circuit the output port and connect a current source I_1 to the input port as shown in Fig. 19.23(a). From Fig. 19.23(a),

$$V_1 = I_1(2 + 3 \parallel 6) = 4I_1$$

Hence,

$$h_{11} = \frac{V_1}{I_1} = 4\ \Omega$$

Also, from Fig. 19.23(a) we obtain, by current division,

$$-I_2 = \frac{6}{6 + 3}I_1 = \frac{2}{3}I_1$$

Hence,

$$h_{21} = \frac{I_2}{I_1} = -\frac{2}{3}$$

—

To obtain h_{12} and h_{22} , we open-circuit the input port and connect a voltage source V_2 to the output port as in Fig. 19.23(b). By voltage division,

$$V_1 = \frac{6}{6 + 3}V_2 = \frac{2}{3}V_2$$

Hence,

$$h_{12} = \frac{V_1}{V_2} = \frac{2}{3}$$

Also,

$$V_2 = (3 + 6)I_2 = 9I_2$$

Hence,

$$h_{22} = \frac{I_2}{V_2} = \frac{1}{9}\text{ S}$$

7. Determine the Thevenin equivalent at the output port of the circuit in Fig. 19.25.

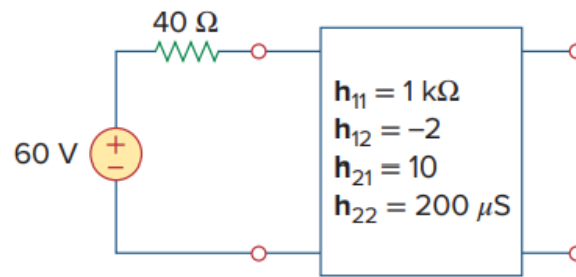


Figure 19.25
For Example 19.6.

Solution:

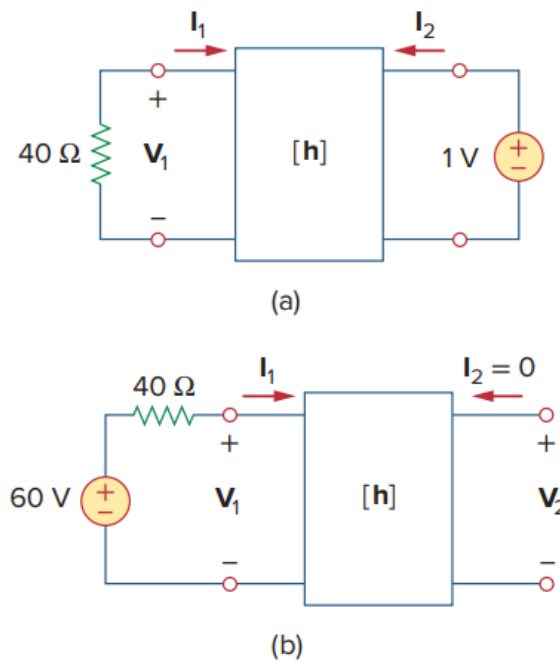


Figure 19.26

For Example 19.6: (a) finding Z_{Th} ,
(b) finding V_{Th} .

To find Z_{Th} and V_{Th} , we apply the normal procedure, keeping in mind the formulas relating the input and output ports of the h model. To obtain Z_{Th} , remove the 60-V voltage source at the input port and apply a 1-V voltage source at the output port, as shown in Fig. 19.26(a). From Eq. (19.14),

$$\mathbf{V}_1 = h_{11}\mathbf{I}_1 + h_{12}\mathbf{V}_2 \quad (19.6.1)$$

$$\mathbf{I}_2 = h_{21}\mathbf{I}_1 + h_{22}\mathbf{V}_2 \quad (19.6.2)$$

But $\mathbf{V}_2 = 1$, and $\mathbf{V}_1 = -40\mathbf{I}_1$. Substituting into Eqs. (19.6.1) and (19.6.2), we get

$$-40\mathbf{I}_1 = h_{11}\mathbf{I}_1 + h_{12} \quad \Rightarrow \quad \mathbf{I}_1 = -\frac{h_{12}}{40 + h_{11}} \quad (19.6.3)$$

$$\mathbf{I}_2 = h_{21}\mathbf{I}_1 + h_{22} \quad (19.6.4)$$

Substituting Eq. (19.6.3) into Eq. (19.6.4),

$$\mathbf{I}_2 = h_{22} - \frac{h_{21}h_{12}}{h_{11} + 40} = \frac{h_{11}h_{22} - h_{21}h_{12} + h_{22} \cdot 40}{h_{11} + 40}$$

Therefore,

$$Z_{Th} = \frac{\mathbf{V}_2}{\mathbf{I}_2} = \frac{1}{\mathbf{I}_2} = \frac{h_{11} + 40}{h_{11}h_{22} - h_{21}h_{12} + h_{22} \cdot 40}$$

Substituting the values of the h parameters,

$$Z_{Th} = \frac{1000 + 40}{10^3 \times 200 \times 10^{-6} + 20 + 40 \times 200 \times 10^{-6}}$$

$$= \frac{1040}{20.21} = 51.46 \Omega$$

—
To get V_{Th} , we find the open-circuit voltage $\mathbf{V}_2$ in Fig. 19.26(b). At the input port,

$$-60 + 40\mathbf{I}_1 + \mathbf{V}_1 = 0 \quad \Rightarrow \quad \mathbf{V}_1 = 60 - 40\mathbf{I}_1 \quad (19.6.5)$$

At the output,

$$\mathbf{I}_2 = 0 \quad (19.6.6)$$

Substituting Eqs. (19.6.5) and (19.6.6) into Eqs. (19.6.1) and (19.6.2), we obtain

$$60 - 40\mathbf{I}_1 = h_{11}\mathbf{I}_1 + h_{12}\mathbf{V}_2 \quad (19.6.7)$$

$$0 = h_{21}\mathbf{I}_1 + h_{22}\mathbf{V}_2 \quad \Rightarrow \quad \mathbf{I}_1 = -\frac{h_{22}}{h_{21}}\mathbf{V}_2 \quad (19.6.8)$$

Now substituting Eq. (19.6.8) into Eq. (19.6.7),

$$60 = \left[-(h_{11} + 40)\frac{h_{22}}{h_{21}} + h_{12} \right] \mathbf{V}_2$$

or

$$V_{Th} = \mathbf{V}_2 = \frac{60}{-(h_{11} + 40)\frac{h_{22}}{h_{21}} + h_{12}} = \frac{60h_{21}}{h_{12}h_{21} - h_{11}h_{22} - 40h_{22}}$$

Substituting the values of the h parameters,

$$V_{Th} = \frac{60 \times 10}{-20.21} = -29.69 \text{ V}$$

8. **Inverse Hybrid parameters** (or g parameters) relate the terminal currents $\mathbf{V}_1, \mathbf{I}_2$ (independent terminal quantities) to the terminal voltages $\mathbf{I}_1, \mathbf{V}_2$ (dependent terminal quantities) as

$$\mathbf{I}_1 = g_{11}\mathbf{V}_1 + g_{12}\mathbf{I}_2$$

$$\mathbf{V}_2 = g_{21}\mathbf{V}_1 + g_{22}\mathbf{I}_2$$

or in matrix form as

$$\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_2 \end{bmatrix} = [g] \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_2 \end{bmatrix}$$

9. Determination of g parameters

To determine g_{11}, g_{21} , we open-circuit the output port ($\mathbf{I}_2 = 0$)

$$g_{11} = \left. \frac{\mathbf{I}_1}{\mathbf{V}_1} \right|_{\mathbf{I}_2=0}, \quad g_{21} = \left. \frac{\mathbf{V}_2}{\mathbf{V}_1} \right|_{\mathbf{I}_2=0}$$

To determine g_{12}, g_{22} , we short-circuit the input port ($\mathbf{V}_1 = 0$)

$$g_{12} = \left. \frac{\mathbf{I}_1}{\mathbf{I}_2} \right|_{\mathbf{V}_1=0}, \quad g_{22} = \left. \frac{\mathbf{V}_2}{\mathbf{I}_2} \right|_{\mathbf{V}_1=0}$$

10. g parameters are frequently used to describe the **field-effect transistors**.

11. Equivalent circuit model

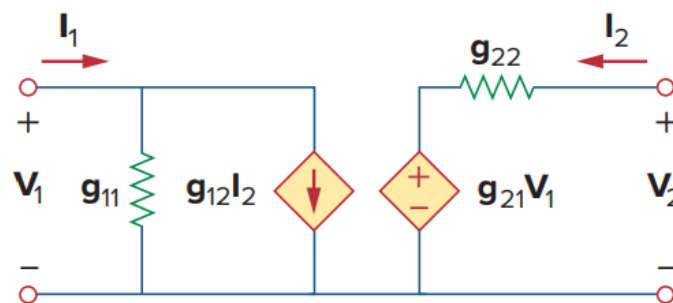


Figure 19.21

The g -parameter model of a two-port network.

12. **Example.** Find the g parameters as functions of s for the circuit in Fig. 19.28.

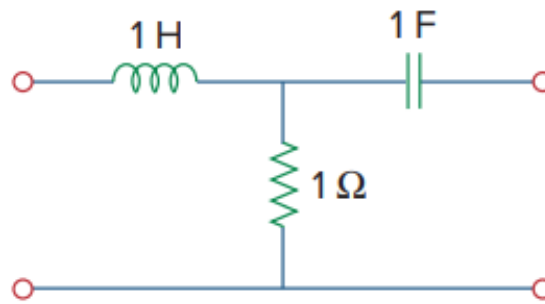
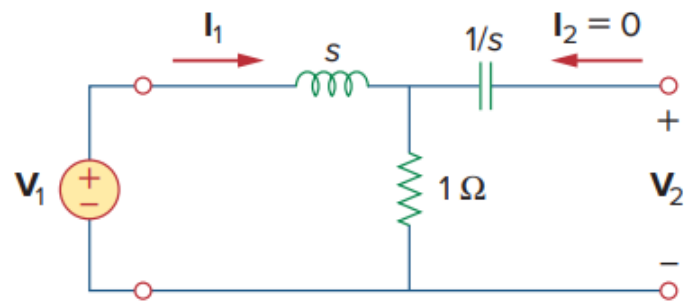
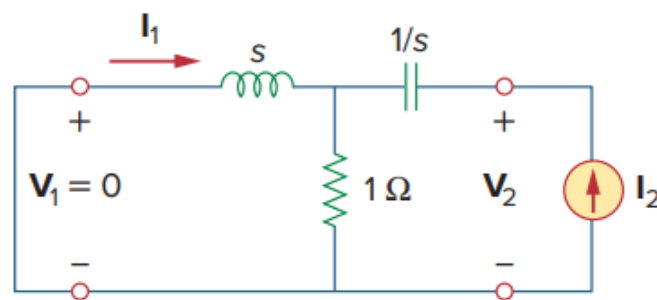


Figure 19.28
For Example 19.7.

Solution: In the s domain,



(a)



(b)

Figure 19.29

Determining the g parameters in the s domain for the circuit in Fig. 19.28.

$$1\text{ H} \Rightarrow sL = s, \quad 1\text{ F} \Rightarrow \frac{1}{sC} = \frac{1}{s}$$

To get g_{11} and g_{21} , we open-circuit the output port and connect a voltage source V_1 to the input port as in Fig. 19.29(a). From the figure,

$$I_1 = \frac{V_1}{s + 1}$$

or

$$g_{11} = \frac{I_1}{V_1} = \frac{1}{s + 1}$$

By voltage division,

$$V_2 = \frac{1}{s + 1} V_1$$

or

$$g_{21} = \frac{V_2}{V_1} = \frac{1}{s + 1}$$

—

To obtain g_{12} and g_{22} , we short-circuit the input port and connect a current source $\mathbf{I}_2$ to the output port as in Fig. 19.29(b). By current division,

$$\mathbf{I}_1 = \frac{-1}{s+1} \mathbf{I}_2$$

or

$$g_{12} = \frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{-1}{s+1}$$

Also,

$$\mathbf{V}_2 = \mathbf{I}_2 \left(\frac{1}{s} + s \parallel 1 \right)$$

or

$$g_{22} = \frac{\mathbf{V}_2}{\mathbf{I}_2} = \frac{1}{s} + \frac{s}{s+1} = \frac{s^2 + s + 1}{s(s+1)}$$

Thus,

$$[g] = \begin{bmatrix} \frac{1}{s+1} & \frac{-1}{s+1} \\ \frac{1}{s+1} & \frac{s^2 + s + 1}{s(s+1)} \end{bmatrix}$$

1.4 Transmission parameters

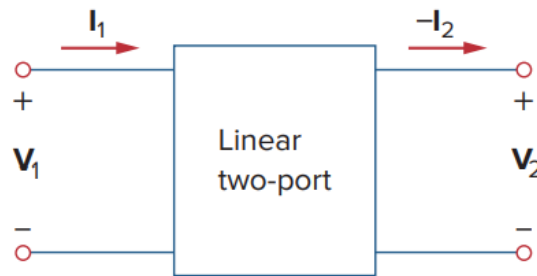


Figure 19.31
Terminal variables used to define the
ADCB parameters.

1. **Transmission parameters** relate the terminal currents $\mathbf{V}_2$, $-\mathbf{I}_2$ (independent terminal quantities) to the terminal voltages $\mathbf{V}_1$, $\mathbf{I}_1$ (dependent terminal quantities) as

$$\begin{aligned}\mathbf{V}_1 &= A\mathbf{V}_2 - B\mathbf{I}_2, \\ \mathbf{I}_1 &= C\mathbf{V}_2 - D\mathbf{I}_2\end{aligned}$$

or in matrix form as

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \mathbf{V}_2 \\ -\mathbf{I}_2 \end{bmatrix} = [T] \begin{bmatrix} \mathbf{V}_2 \\ -\mathbf{I}_2 \end{bmatrix}$$

2. The transmission parameters provide a measure of how a circuit transmits voltage and current from a source to a load. They are useful in the analysis of transmission lines (such as cable and fiber) because they express sending-end variables ($\mathbf{V}_1$ and $\mathbf{I}_1$) in terms of the receiving-end variables ($\mathbf{V}_2$ and $-\mathbf{I}_2$).
3. The transmission parameters are also known as **ABCD parameters**. They are used in the design of telephone systems, microwave networks, and radars.
4. **Determination of transmission parameters**

$$\begin{aligned}A &= \left. \frac{\mathbf{V}_1}{\mathbf{V}_2} \right|_{\mathbf{I}_2=0}, & B &= - \left. \frac{\mathbf{V}_1}{\mathbf{I}_2} \right|_{\mathbf{V}_2=0}, \\ C &= \left. \frac{\mathbf{I}_1}{\mathbf{V}_2} \right|_{\mathbf{I}_2=0}, & D &= - \left. \frac{\mathbf{I}_1}{\mathbf{I}_2} \right|_{\mathbf{V}_2=0}\end{aligned}$$

5. **Inverse transmission parameters** express the variables at the output port $\mathbf{V}_2, \mathbf{I}_2$ (dependent terminal quantities) in terms of the variables at the input port $\mathbf{V}_1, -\mathbf{I}_1$ (independent terminal quantities) as

$$\mathbf{V}_2 = a\mathbf{V}_1 - b\mathbf{I}_1, \quad \mathbf{I}_2 = c\mathbf{V}_1 - d\mathbf{I}_1$$

or in matrix form as

$$\begin{bmatrix} \mathbf{V}_2 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ -\mathbf{I}_1 \end{bmatrix} = [t] \begin{bmatrix} \mathbf{V}_1 \\ -\mathbf{I}_1 \end{bmatrix}$$

6. **Determination of inverse transmission parameters**

$$a = \left. \frac{\mathbf{V}_2}{\mathbf{V}_1} \right|_{\mathbf{I}_1=0}, \quad b = - \left. \frac{\mathbf{V}_2}{\mathbf{I}_1} \right|_{\mathbf{V}_1=0},$$

$$c = \left. \frac{\mathbf{I}_2}{\mathbf{V}_1} \right|_{\mathbf{I}_1=0}, \quad d = - \left. \frac{\mathbf{I}_2}{\mathbf{I}_1} \right|_{\mathbf{V}_1=0}$$

7. In terms of the transmission or inverse transmission parameters, a network is **reciprocal** if

$$\mathbf{AD} - \mathbf{BC} = 1, \quad \mathbf{ad} - \mathbf{bc} = 1$$

8. **Example.** Find the transmission parameters for the two-port network in Fig. 19.32.

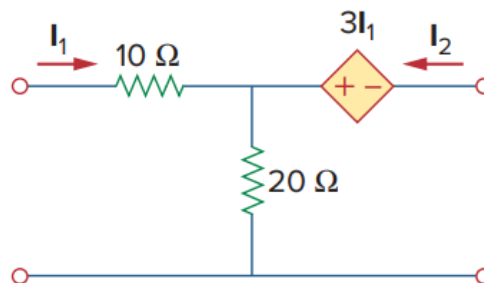


Figure 19.32
For Example 19.8.

Solution:

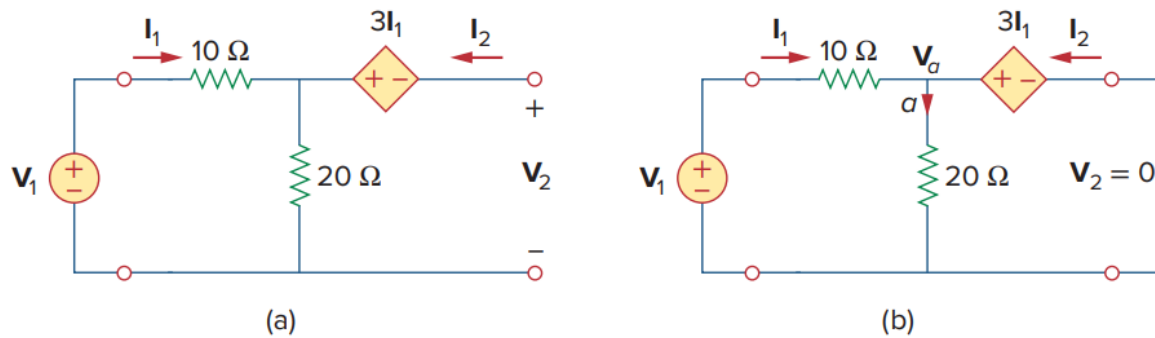


Figure 19.33

For Example 19.8: (a) finding **A** and **C**, (b) finding **B** and **D**.

To determine A and C , we leave the output port open as in Fig. 19.33(a) so that $I_2 = 0$ and place a voltage source V_1 at the input port. We have

$$V_1 = (10 + 20)I_1 = 30I_1, \quad V_2 = 20I_1 - 3I_1 = 17I_1$$

Thus,

$$A = \frac{V_1}{V_2} = \frac{30I_1}{17I_1} = 1.765, \quad C = \frac{I_1}{V_2} = \frac{I_1}{17I_1} = 0.0588 \text{ S}$$

—

To obtain B and D , we short-circuit the output port so that $V_2 = 0$ as shown in Fig. 19.33(b) and place a voltage source V_1 at the input port. At node a in the circuit of Fig. 19.33(b), KCL gives

$$\frac{V_1 - V_a}{10} - \frac{V_a}{20} + I_2 = 0 \quad (19.8.1)$$

But $V_a = 3I_1$ and $I_1 = (V_1 - V_a)/10$. Combining these gives

$$V_a = 3I_1, \quad V_1 = 13I_1 \quad (19.8.2)$$

Substituting $V_a = 3I_1$ into Eq. (19.8.1) and replacing the first term with I_1 ,

$$I_1 - \frac{3I_1}{20} + I_2 = 0 \quad \Rightarrow \quad \frac{17}{20}I_1 = -I_2$$

Therefore,

$$D = \frac{I_1}{I_2} = \frac{20}{17} = 1.176, \quad B = \frac{V_1}{I_2} = \frac{-13I_1}{(-17/20)I_1} = 15.29 \Omega$$

Example.

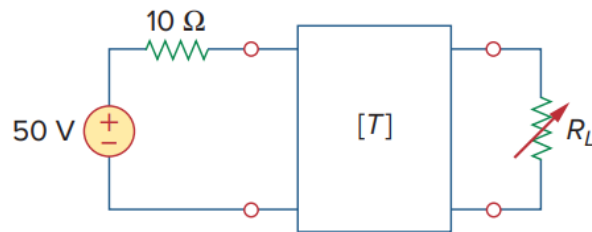


Figure 19.34
For Example 19.9.

The **ABCD** parameters of the two-port network in Fig. 19.34 are

$$\begin{bmatrix} 4 & 20\ \Omega \\ 0.1\ \text{S} & 2 \end{bmatrix}$$

The output port is connected to a variable load for maximum power transfer. Find R_L and the maximum power transferred.

Solution: What we need is to find the Thevenin equivalent (Z_{Th} and V_{Th}) at the

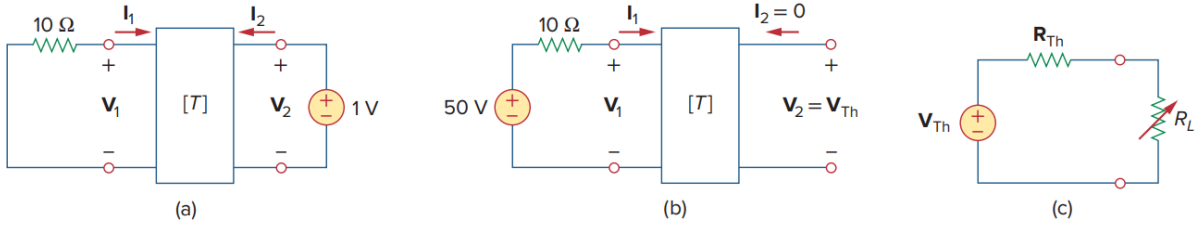


Figure 19.35

Solution of Example 19.9: (a) finding Z_{Th} , (b) finding V_{Th} , (c) finding R_L for maximum power transfer.

load or output port. We find Z_{Th} using the circuit in Fig. 19.35(a). Our goal is to get $Z_{Th} = \frac{V_2}{I_2}$.

Substituting the given **ABCD** parameters into Eq. (19.22), we obtain

$$V_1 = 4V_2 - 20I_2 \quad (19.9.1)$$

$$I_1 = 0.1V_2 - 2I_2 \quad (19.9.2)$$

At the input port, $V_1 = -10I_1$. Substituting this into Eq. (19.9.1) gives

$$-10I_1 = 4V_2 - 20I_2$$

or

$$I_1 = -0.4V_2 + 2I_2 \quad (19.9.3)$$

Setting the right-hand sides of Eqs. (19.9.2) and (19.9.3) equal,

$$0.1V_2 - 2I_2 = -0.4V_2 + 2I_2 \Rightarrow 0.5V_2 = 4I_2$$

Hence,

$$Z_{Th} = \frac{V_2}{I_2} = \frac{4}{0.5} = 8 \Omega$$

To find V_{Th} , we use the circuit in Fig. 19.35(b). At the output port $I_2 = 0$ and at the input port $V_1 = 50 - 10I_1$. Substituting these into Eqs. (19.9.1) and (19.9.2),

$$50 - 10I_1 = 4V_2 \quad (19.9.4)$$

$$I_1 = 0.1V_2 \quad (19.9.5)$$

Substituting Eq. (19.9.5) into Eq. (19.9.4),

$$50 - V_2 = 4V_2 \Rightarrow V_2 = 10$$

Thus,

$$V_{Th} = V_2 = 10 V$$

The equivalent circuit is shown in Fig. 19.35(c). For maximum power transfer,

$$R_L = Z_{Th} = 8 \Omega$$

From Eq. (4.24), the maximum power is

$$P = I^2 R_L = \left(\frac{V_{Th}}{2R_L} \right)^2 R_L = \frac{V_{Th}^2}{4R_L} = \frac{100}{4 \times 8} = 3.125 W$$

1.5 Relationship between 6 parameters

TABLE 19.1

Conversion of two-port parameters.

	z		y		h		g		T		t	
z	z_{11}	z_{12}	$\frac{y_{22}}{\Delta_y}$	$-\frac{y_{12}}{\Delta_y}$	$\frac{\Delta_h}{h_{22}}$	$\frac{h_{12}}{h_{22}}$	$\frac{1}{g_{11}}$	$-\frac{g_{12}}{g_{11}}$	$\frac{A}{C}$	$\frac{\Delta_T}{C}$	$\frac{d}{c}$	$\frac{1}{c}$
	z_{21}	z_{22}	$-\frac{y_{21}}{\Delta_y}$	$\frac{y_{11}}{\Delta_y}$	$-\frac{h_{21}}{h_{22}}$	$\frac{1}{h_{22}}$	$\frac{g_{21}}{g_{11}}$	$\frac{\Delta_g}{g_{11}}$	$\frac{1}{C}$	$\frac{D}{C}$	$\frac{\Delta_t}{c}$	$\frac{a}{c}$
y	$\frac{z_{22}}{\Delta_z}$	$-\frac{z_{12}}{\Delta_z}$	y_{11}	y_{12}	$\frac{1}{h_{11}}$	$-\frac{h_{12}}{h_{11}}$	$\frac{\Delta_g}{g_{22}}$	$\frac{g_{12}}{g_{22}}$	$\frac{D}{B}$	$-\frac{\Delta_T}{B}$	$\frac{a}{b}$	$\frac{1}{b}$
	$-\frac{z_{21}}{\Delta_z}$	$\frac{z_{11}}{\Delta_z}$	y_{21}	y_{22}	$\frac{h_{21}}{h_{11}}$	$\frac{\Delta_h}{h_{11}}$	$-\frac{g_{21}}{g_{22}}$	$\frac{1}{g_{22}}$	$-\frac{1}{B}$	$\frac{A}{B}$	$-\frac{\Delta_t}{b}$	$\frac{d}{b}$
h	$\frac{\Delta_z}{z_{22}}$	$\frac{z_{12}}{z_{22}}$	$\frac{1}{y_{11}}$	$-\frac{y_{12}}{y_{11}}$	h_{11}	h_{12}	$\frac{g_{22}}{\Delta_g}$	$-\frac{g_{12}}{\Delta_g}$	$\frac{B}{D}$	$\frac{\Delta_T}{D}$	$\frac{b}{a}$	$\frac{1}{a}$
	$-\frac{z_{21}}{z_{22}}$	$\frac{1}{z_{22}}$	$\frac{y_{21}}{y_{11}}$	$\frac{\Delta_y}{y_{11}}$	h_{21}	h_{22}	$-\frac{g_{21}}{\Delta_g}$	$\frac{g_{11}}{\Delta_g}$	$-\frac{1}{D}$	$\frac{C}{D}$	$\frac{\Delta_t}{a}$	$\frac{c}{a}$
g	$\frac{1}{z_{11}}$	$-\frac{z_{12}}{z_{11}}$	$\frac{\Delta_y}{y_{22}}$	$\frac{y_{12}}{y_{22}}$	$\frac{h_{22}}{\Delta_h}$	$-\frac{h_{12}}{\Delta_h}$	g_{11}	g_{12}	$\frac{C}{A}$	$-\frac{\Delta_T}{A}$	$\frac{c}{d}$	$\frac{1}{d}$
	$\frac{z_{21}}{z_{11}}$	$\frac{\Delta_z}{z_{11}}$	$-\frac{y_{21}}{y_{22}}$	$\frac{1}{y_{22}}$	$-\frac{h_{21}}{\Delta_h}$	$\frac{h_{11}}{\Delta_h}$	g_{21}	g_{22}	$\frac{1}{A}$	$\frac{B}{A}$	$\frac{\Delta_t}{d}$	$\frac{b}{d}$
T	$\frac{z_{11}}{z_{21}}$	$\frac{\Delta_z}{z_{21}}$	$-\frac{y_{22}}{y_{21}}$	$\frac{1}{y_{21}}$	$-\frac{\Delta_h}{h_{21}}$	$-\frac{h_{11}}{h_{21}}$	$\frac{1}{g_{21}}$	$\frac{g_{22}}{g_{21}}$	A	B	$\frac{d}{\Delta_t}$	$\frac{b}{\Delta_t}$
	$\frac{1}{z_{21}}$	$\frac{z_{22}}{z_{21}}$	$-\frac{\Delta_y}{y_{21}}$	$\frac{y_{11}}{y_{21}}$	$-\frac{h_{22}}{h_{21}}$	$\frac{1}{h_{21}}$	$\frac{g_{11}}{g_{21}}$	$\frac{\Delta_g}{g_{21}}$	C	D	$\frac{c}{\Delta_t}$	$\frac{a}{\Delta_t}$
t	$\frac{z_{22}}{z_{12}}$	$\frac{\Delta_z}{z_{12}}$	$-\frac{y_{11}}{y_{12}}$	$\frac{1}{y_{12}}$	$\frac{1}{h_{12}}$	$\frac{h_{11}}{h_{12}}$	$-\frac{\Delta_g}{g_{12}}$	$-\frac{g_{22}}{g_{12}}$	$\frac{D}{\Delta_T}$	$\frac{B}{\Delta_T}$	a	b
	$\frac{1}{z_{12}}$	$\frac{z_{11}}{z_{12}}$	$-\frac{\Delta_y}{y_{12}}$	$\frac{y_{22}}{y_{12}}$	$\frac{h_{22}}{h_{12}}$	$\frac{\Delta_h}{h_{12}}$	$\frac{g_{11}}{g_{12}}$	$-\frac{1}{g_{12}}$	$\frac{C}{\Delta_T}$	$\frac{A}{\Delta_T}$	c	d

$$\Delta_z = z_{11}z_{22} - z_{12}z_{21},$$

$$\Delta_y = y_{11}y_{22} - y_{12}y_{21},$$

$$\Delta_h = h_{11}h_{22} - h_{12}h_{21},$$

$$\Delta_g = g_{11}g_{22} - g_{12}g_{21},$$

$$\Delta_T = AD - BC$$

$$\Delta_t = ad - bc$$

1.6 Interconnection of two-port networks

1.6.1 Series connection

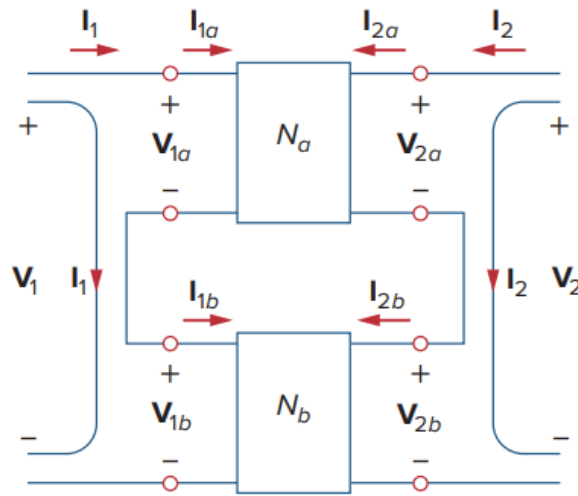


Figure 19.39

Series connection of two two-port networks.

1.6.2 Parallel connection

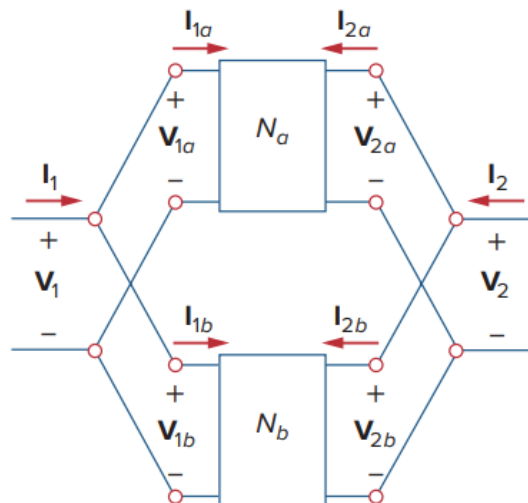
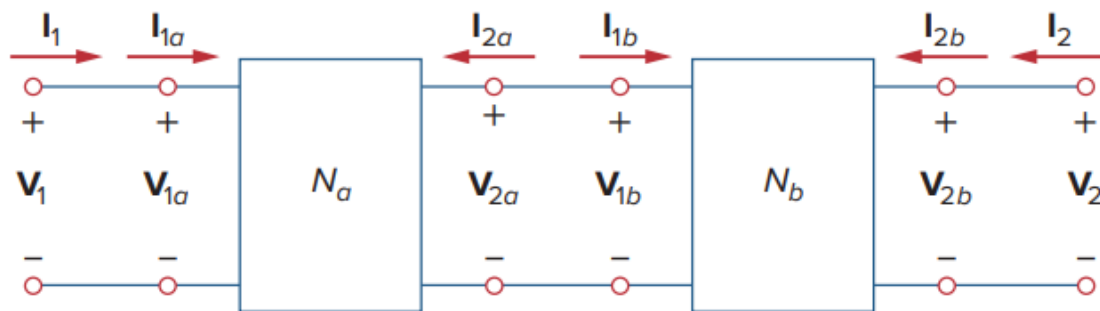


Figure 19.40

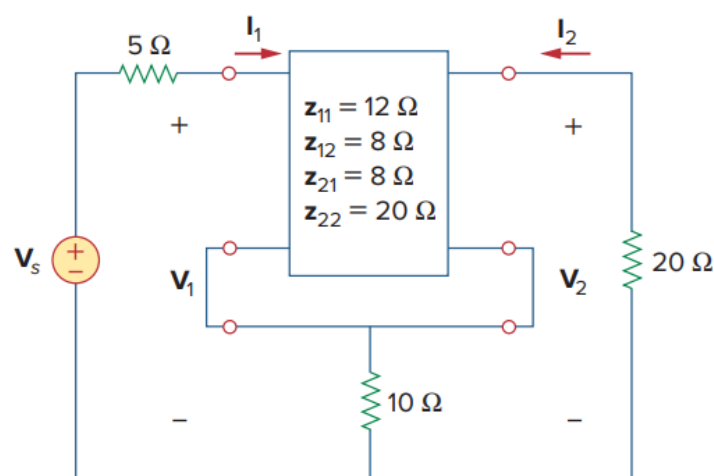
Parallel connection of two two-port networks.

1.6.3 Cascaded connection

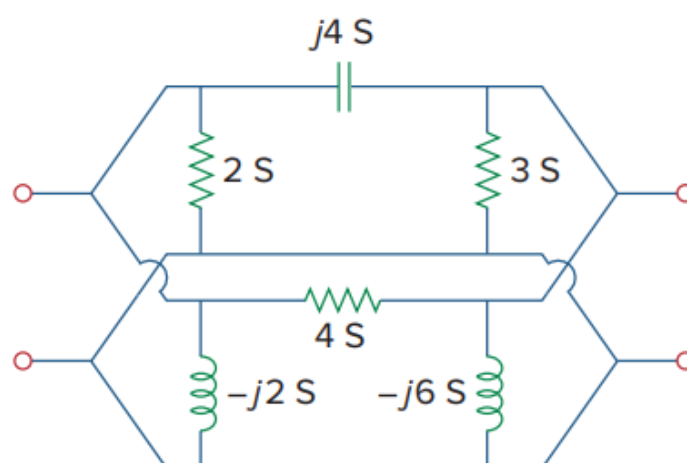
1.6.4 Examples.

**Figure 19.41**

Cascade connection of two two-port networks.

**Figure 19.42**

For Example 19.12.

**Figure 19.44**

For Example 19.13.

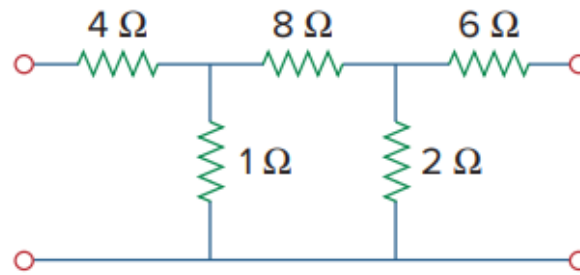
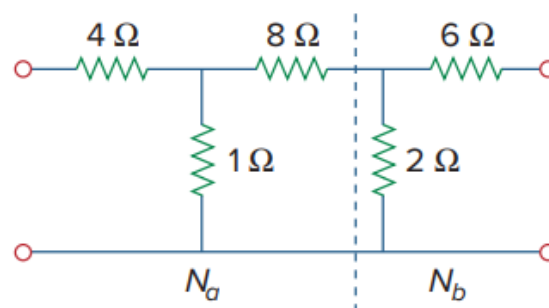
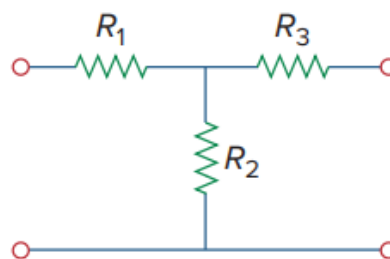


Figure 19.46
For Example 19.14.



(a)



(b)

Figure 19.47
For Example 19.14: (a) Breaking the
circuit in Fig. 19.46 into two two-ports,
(b) a general T two-port.